Introduction to Data Structures and Algorithms

Dr. Sirasit Lochanachit

Outline

1. Resources
2. What is a “Data Structure”?
3. What is an “Algorithm”?
4. Prerequisites
5. Lesson Plan
6. Grading

Resources

1. Course Materials Website:

   https://github.com/noswolf/DSA_BIT/tree/DSA_24

2. Google Colaboratory or VSCode (offline) for Labs

   - Interactive notebooks (.ipynb)
   - https://colab.research.google.com/
   - Login with your kmitl email address

What is a "Data Structure" ?

How do we store, organise, and retrieve data on a computer?

Abstract Data Type

- A data type where only **behavior** is defined but not implementation.
- Examples: Array, List, Map, Queue, Set, and etc.

Common vs Abstract Data Type

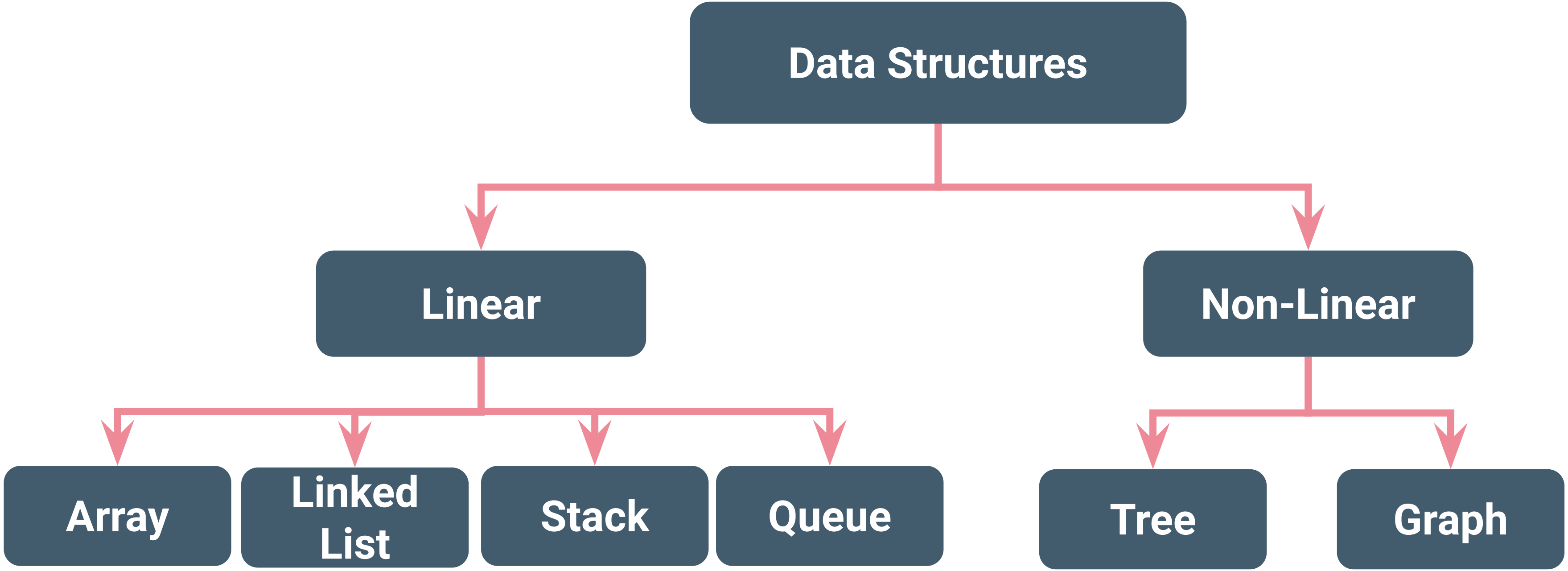
Common

- Integer
- Floating-point number
- Character
- String
- Boolean
- etc.

Abstract

- Array
- List
- Map
- Queue
- etc.

Type of Data Structure



Check out for a comprehensive list of data structures at
https://en.wikipedia.org/wiki/List_of_data_structures

What is an “Algorithm” ?

Why care about an “Algorithm” ?

How can we efficiently (in space/time) carry out some typical data processing operations?

How do we analyze and describe their performance?

Example: Sorting numbers

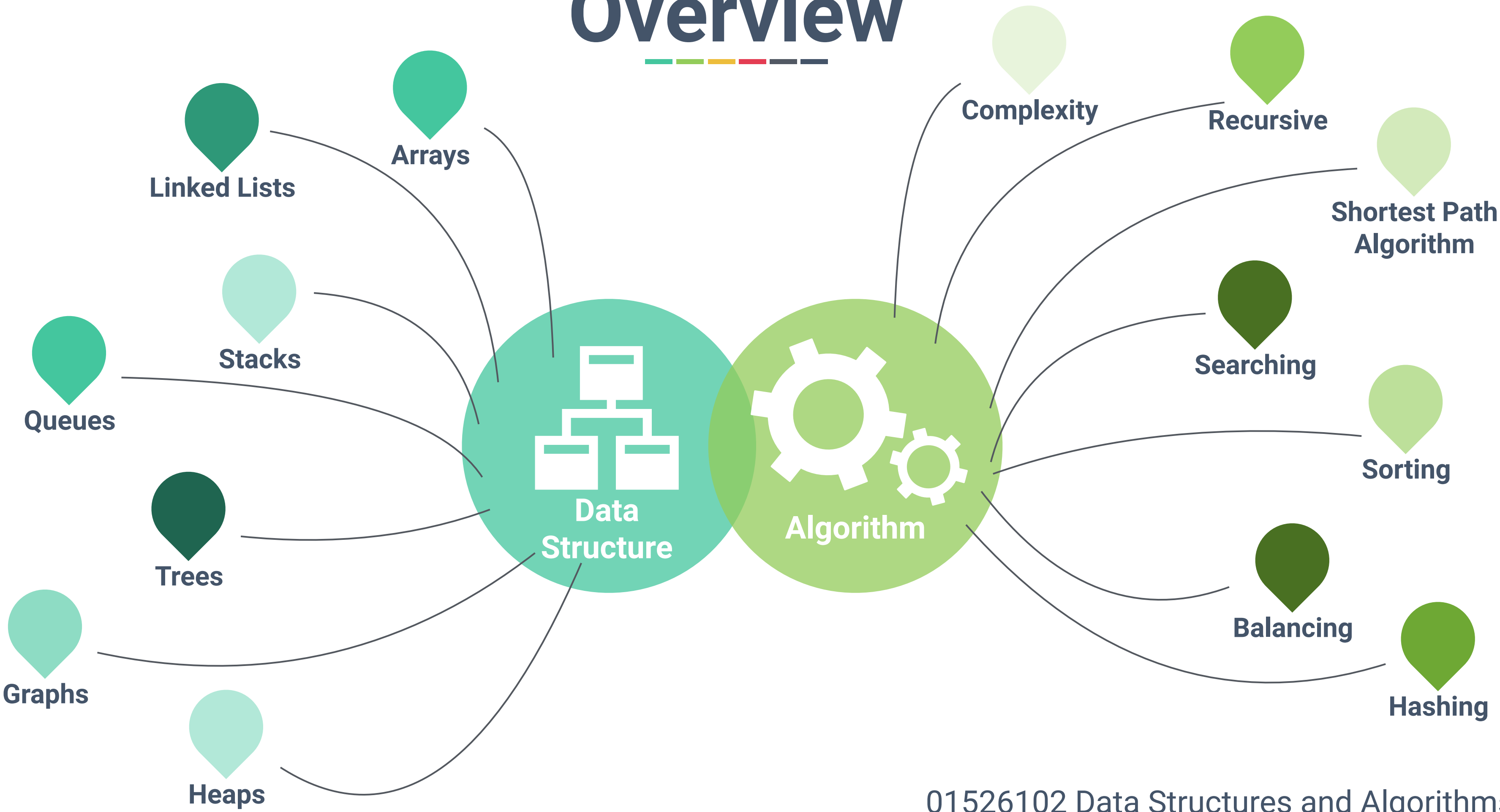
1. Input:
   - A sequence of n numbers: $<a_1, a_2, \ldots, a_n>$
   - <31, 41, 59, 26, 41>
2. Sorting Algorithms
3. Output:
   - A permutation (reordering) $<a'_1, a'_2, \ldots, a'_n>$ of input sequence such that $a'_1 <= a'_2 <= \ldots <= a'_n$
   - <26, 31, 41, 41, 59>

What kind of problems are solved by algorithms?

- Human Genome Project
  - identifying all genes of human beings
- Internet: Routing, searches, and security
  - **Shortest path**, search engines, encrypted communication
- E-commerce
  - Ads, recommendations, authentications
- Commercial enterprises
  - Resource allocation:
    - crew assignment on flights, package delivery route

SCHOOL OF INTERNATIONAL & INTERDISCIPLINARY ENGINEERING PROGRAMS
King Mongkut's Institute of Technology Ladkrabang

Overview



Summary

- Data Structure
    - Way to store and organise data, allowing operations to be performed efficiently.
- Algorithm
    - Step-by-step procedure, which performs on data structure, to be followed to solve a problem/accomplish a task.

Prerequisites

- Fluent in Python Programming
- Comfortable with development processes
  - Writing a function
  - Debugging and testing a code

Lesson Plan (till Midterm)

*Public holidays are highlighted in red

Week	Topics	Labs
25/11/2024	Introduction	#1
02/12/2024	Algorithm Analysis	#2
09/12/2024	Arrays	#3
16/12/2024	Stacks	#4
23/12/2024	Queues	
30/12/2024	-	-
06/01/2025	Linked Lists	#5
13/01/2025	Linked Lists (cont.)	#6

Lesson Plan (after Midterm)

*Public holidays are highlighted in red

Week	Topics	Labs
27/01/2025	Binary Search & Hash Tables	#7
03/02/2025	Sorting	#8
10/02/2025	Recursion	
17/02/2025	Trees	#9
24/02/2025	Binary Search Trees	
03/03/2025	Graphs	#10
10/03/2025	-	-
19/03/2025	Final Exam	

Grading

Attendance	5%
Lab Assignment	25%
Midterm Exam	35%
Final Exam	35%

Attendance Score

Out of 13 times, If you attend onsite

12+ times = 5%

11 times = 4%

10 times = 2%

9 times = Unable to take a final exam

Note: Leave notice must be submitted at least <u>48 hrs</u> before the lecture.

Note 2: Lab submission != Checked Attendance

Lab Assignment

Each lab required to be submitted and presented to TAs during the lab session only

- Late submission is not allowed and will not be graded
- Your TAs will be your grader, not your instructor
- If you need any help, feel free to ask me or your TAs
- MS Teams submission is required for your own benefit.
- You can ask your instructor to see a checklist of your submissions.
- Turn off AI generative features in your Colab environment

Suggestion

Working on Lab tasks

- Although this course is intense on programming, you are encouraged to work on lab with your classmates, including having intense discussion.
- If you are stuck, unsure about the task, or have no idea about the error message, feel free to ask TA and they are happy to help.

Suggestion

Generative AI age

- Even though AI features are allowed and available for you to use freely in this course, please restrain from using it as a first thing when you attempting any task.
- It can be very useful if you run out of idea/solution or need a hint from it, including help prepare you for the exam.

Things to Seriously Consider

- **Recording** my lectures in any capacity **is not allowed**.

- However, you are allowed to
  - Take photo of my handwritings on the whiteboard or
  - If you would like my VDO lectures, feel free to ask me after the lecture session.
- Alternatively, Public lectures on YouTube is a great material for you to understand the complex data structures concepts.

Reading List

Essential

Goodrich, M.T., Tamassia, R. and Goldwasser, M.H., 2013. ***Data structures and algorithms in Python***. John Wiley & Sons Ltd.

Recommended

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2022. ***Introduction to algorithms***. MIT press.

Miller, B.N. and Ranum, D.L., 2011. ***Problem solving with algorithms and data structures using python***, 2nd ed. Franklin, Beedle & Associates Inc.